

Rolling Disk on an Inclined Plane

Dynamics, Energy, and Transition Analysis

Group 5

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Introduction

A disk on a slope can either roll smoothly or slip. This project explains the physics behind both behaviors

Key Tasks:

- Derive equations for rolling and slipping, determine transition condition
- Simulate and visualize velocity, angular velocity, friction forces, and energy distribution

Definitions:

Translational Motion: Motion of the disk's center moving down the incline (linear velocity v).

Rotational Motion: Spinning motion of the disk about its center (angular velocity ω).

Transition: The moment when the disk switches from slipping to rolling, when the no-slip condition $v = \omega r$ is satisfied.

System Description: Disk of mass m , radius r , on a plane inclined at angle θ

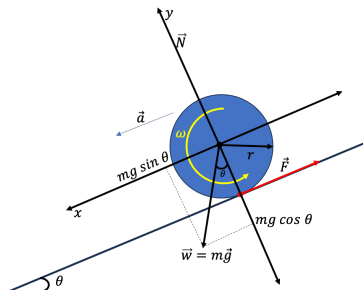


Figure: Rolling Disk on Inclined Plane, *Research Gate 2025* .

Variables:

- v : linear velocity (down incline)
- ω : angular velocity (rotation)
- F : friction force (static or kinetic)
- θ : incline angle

Disk Moment of Inertia

For a uniform disk:

$$I = \frac{1}{2}mr^2$$

Interpretation:

- Depends on mass distribution (all mass at equal distance from axis)
- Key parameter in rotational dynamics
- Affects how friction translates to angular acceleration

Energy in Rotation:

$$KE_{\text{rot}} = \frac{1}{2}I\omega^2 = \frac{1}{4}mr^2\omega^2$$

Pure Rolling: Regime 1

No-Slip Condition: The contact point between disk and slope is momentarily at rest, that is, contact point velocity is zero

$$v = \omega r$$

Differentiating:

$$\ddot{x} = \alpha r$$

According to Newton's second Law of Motion ($\sum \mathbf{F} = m\mathbf{a}$):

$$m\ddot{x} = mg \sin \theta - f_s \quad (1)$$

Torque Balance (about center of mass):

$$I\alpha = f_s r \quad (2)$$

where f_s is static friction (unknown, to be determined)

Pure Rolling: Equations of Motion

From torque equation (2):

$$f_s = \frac{I\alpha}{r} = \frac{\frac{1}{2}mr^2 \cdot \alpha}{r} = \frac{1}{2}mr\alpha = \frac{1}{2}m\ddot{x}$$

Substitute into force equation (1):

$$m\ddot{x} = mg \sin \theta - \frac{1}{2}m\ddot{x}$$

$$\frac{3}{2}m\ddot{x} = mg \sin \theta$$

$$\ddot{x}_{roll} = \frac{2}{3}g \sin \theta$$

Friction force:

$$f_s = \frac{1}{3}mg \sin \theta$$

Angular acceleration:

$$\alpha = \frac{2g \sin \theta}{3r}$$

Slipping: Regime 2

No constraint $v \neq \omega r$, that is, v and ω evolve independently during slipping.

Force and Torque Equations:

$$m\ddot{x} = mg \sin \theta - f_k$$

$$I\alpha = f_k r$$

where $f_k = \mu_k N = \mu_k mg \cos \theta$ (kinetic friction)

Linear Acceleration:

$$\ddot{x}_{\text{slip}} = g \sin \theta - \mu_k g \cos \theta$$

Angular Acceleration:

$$\alpha_{\text{slip}} = \frac{\mu_k mg \cos \theta \cdot r}{\frac{1}{2}mr^2} = \frac{2\mu_k g \cos \theta}{r}$$

Transition from Slipping to Rolling

Slipping ends when the no-slip condition is naturally satisfied:

$$v(t^*) = \omega(t^*)r$$

Starting from rest with initial angular velocity ω_0 :

- Linear Velocity: $v(t) = (g \sin \theta - \mu_k g \cos \theta)t$
- Angular velocity: $\omega(t) = \omega_0 + \frac{2\mu_k g \cos \theta}{r}t$

Transition time:

$$t^* = \frac{\omega_0 r}{g(\sin \theta - 3\mu_k \cos \theta)}$$

After transition: System follows pure rolling dynamics with

$$a_{\text{roll}} = \frac{2}{3}g \sin \theta$$

Energy Distribution: Pure Rolling

Total Kinetic Energy:

$$KE_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

Using $v = \omega r$ and $I = \frac{1}{2}mr^2$:

$$KE_{\text{trans}} = \frac{1}{2}mv^2$$

$$KE_{\text{rot}} = \frac{1}{2} \cdot \frac{1}{2}mr^2 \cdot \frac{v^2}{r^2} = \frac{1}{4}mv^2$$

$$KE_{\text{total}} = \frac{3}{4}mv^2$$

Energy Ratio:

- Translational: $\frac{2}{3}$ of total
- Rotational: $\frac{1}{3}$ of total

Energy Distribution: Slipping and Rolling

Total Kinetic Energy:

$$KE_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

Using $I = \frac{1}{2}mr^2$ and $v \neq \omega r$:

$$KE_{\text{trans}} = \frac{1}{2}mv^2$$

$$KE_{\text{rot}} = \frac{1}{2} \cdot \frac{1}{2}mr^2 \cdot \omega^2 = \frac{1}{4}mr^2\omega^2$$

$$KE_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{4}mr^2\omega^2$$

Energy Loss (Friction Dissipation):

$$E_{\text{loss}} = \mu_k mg \cos \theta \times d_{\text{slip}}$$

where d_{slip} is the sliding distance at contact point.

Key Difference: No fixed ratio. Translational and rotational energies evolve independently and depend on the relative velocity at contact.

Regimes: Static vs Kinetic Friction

Property	Pure Rolling	Slipping
Friction Type	Static f_s	Kinetic f_k
Magnitude	$\frac{1}{3}mg \sin \theta$	$\mu_k mg \cos \theta$
Constraint	$f_s < \mu_s mg \cos \theta$	$f_k = \mu_k mg \cos \theta$
Energy Loss	None (contact at rest)	Yes (relative motion)
Work by Friction	$W_s = 0$ (no displacement)	$W_k = \mu_k mg \cos \theta \times d_{\text{slip}}$

Condition for Pure Rolling:

$$\tan \theta < 3\mu_s \text{ approximately}$$

If $\tan \theta$ is large enough, the disk immediately slips

Simulation link (<https://inclined-disk.onrender.com/>)

Simulation: The 2 velocities in the 2 Regimes

Scenario: Disk starts from rest, $\theta = 45^\circ$, $\mu_k = 0.1$, $\mu_s = 0.15$

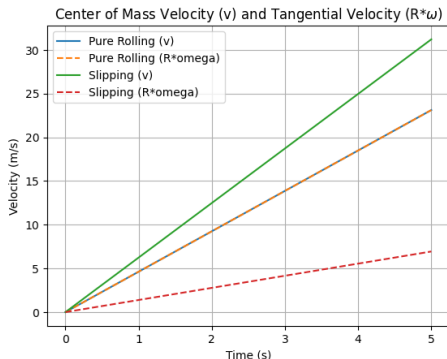


Figure: Linear Velocity Vs Angular Velocity

Insight: The disk slides down faster than it spins, then naturally settles into smooth rolling once linear and rotational motion match.

1. Rolling is Efficient:

- Static friction does no work (contact point at rest)
- Energy is conserved; gravity purely converts to kinetic energy

2. Slipping Dissipates Energy:

- Kinetic friction acts on moving contact point
- Energy is lost as heat; disk accelerates less than on frictionless surface

3. Moment of Inertia Matters:

- Disk: $I = \frac{1}{2}mr^2$ (moderate rotational inertia)
- Larger I means more energy goes into rotation

4. Incline Angle Effect:

- Steeper incline $\rightarrow$ faster acceleration but higher slipping tendency
- Shallow incline $\rightarrow$ stable rolling, pure rolling always occurs

Applications & Conclusion

Key Results:

- Pure rolling: $a = \frac{2}{3}g \sin \theta$, friction $f_s = \frac{1}{3}mg \sin \theta$
- Slipping: $a = g(\sin \theta - \mu_k \cos \theta)$, friction $f_k = \mu_k mg \cos \theta$
- Energy splits: $\frac{2}{3}$ translational, $\frac{1}{3}$ rotational
- Transition occurs when $v = \omega r$ constraint is satisfied

Real-World Applications:

- Vehicles on inclines: braking and traction control
- Train : wheel- rail contact
- Ball bearings and machinery: efficiency optimization
- Road vehicle safety : tires on road

Future Perspectives :

- Generalization to other shapes (sphere, ring, hollow)
- Train : effect of air resistance
- Non-Uniform surfaces (variable friction)

Thank You!